

Water Quality Mapping of Yamuna River Stretch Passing Through Delhi State Using High Resolution Geoeye-2 Imagery

Saif Said, Civil Engineering Department, Aligarh Muslim University, Aligarh, India

Athar Hussain, Civil Engineering Department, Ch. Bhrum Prakash College of Engineering, New Delhi, India

Garima Sharma, Gautam Buddh University, Greater Noida, India

ABSTRACT

The present article utilizes high resolution Geoeye 2 imagery for mapping and monitoring pollution concentrations of 22 km stretch of river Yamuna passing through Delhi state, by developing regression models between water quality parameters (WQP's) and the corresponding spectral reflectance values. Water samples collected from the sampling locations were analysed for 20 WQP's and grouped into four classes namely; (WQP)_{organic}, (WQP)_{inorganic}, (WQP)_{anion} and (WQP)_{cation}. Several spectral band combinations as well as single bands were probed for performing multiple linear regression (MLR) analysis with the four WQP classes. Results reveal relatively strong positive correlations for band combination viz. $[\text{mean RGB} \times \sqrt{B/R}]$ with all four WQP classes yielding high R^2 value (~ 0.85) and RMSE (~ 1.03) amongst other selected band combinations. Spatial distribution maps were generated that substantiates to the actual in-situ pollution concentration levels thereby evidences the potential of high resolution Geoeye-2 imagery for monitoring and mapping pollution concentrations in the water bodies.

KEYWORDS

Digital Number (DN), Multiple Linear Regression (MLR), Point and Non Point Sources, Spectral Reflectance, Water Quality Parameters (WQP's), Yamuna River,

INTRODUCTION

River Yamuna, the largest tributary of river Ganges originates from Yamunotri glacier at an elevation of 6387 m, at the Bander poonch peak in the Uttarkashi district of Uttarakhand. It serves as the primary source of drinking water for Delhi and other neighboring states of Uttar Pradesh, Haryana and Uttarakhand. Last few decades have witnessed severe deterioration in its water quality especially within the 22 km long stretch passing through Delhi state wherein significant water abstraction at Wazirabad barrage (entry point), ITO barrage (midway) and Okhla barrage (exit point) reduces the dilution capacity of the segment. The Delhi segment of river Yamuna is considered as the most polluted stretch since it receives wastewater from seventeen sewage drains of Delhi besides Najafgarh drain.

Determination of water parameters is used for description of the quality of water. Tests for physical/chemical parameters monitor the characteristics that affect the appearance, taste and odor of water but generally do not cause a health risk, while microbiological tests are for water borne organisms that could potentially cause disease. The quality of water is defined by its physical, biological and chemical compositions where chemical composition includes the organic and inorganic substances

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such as heavy metals, pesticides, detergents and petroleum. The physical composition comprises of turbidity, color and temperature, whereas the biological composition includes plankton and pigment (Allee and Jhonson, 1999).

In addition, some of the parameters that defines the criteria for healthy river water and are widely being analyzed in water quality based studies include pH, turbidity, dissolved oxygen (DO), total suspended solids (TSS), total dissolved solids (TDS), turbidity, total hardness as CaCO_3 , biochemical oxygen demand (BOD), chemical oxygen demand (COD), calcium (Ca), magnesium (Mg), alkalinity, phosphate (PO_4), sodium (Na), potassium (K), sulphate (SO_4) and nitrate (NO_3) to name a few. Monitoring and assessment of these water quality parameters requires sampling from widely distributed locations which is time consuming and requires a lot of field and laboratory efforts to present statistical results (Singh, et al., 2013; Duong N. Dinh, 2012; Amandeep V., 2011; Kazi et al., 2009; Ekercin, 2007; Icaga, 2007; Wang et al., 2004; Silvert, 1998; Pattiaratchi et al., 1994). However, remote sensing in conjunction with Geographic Information System (GIS) provide better estimates and synoptic coverage of spatial distribution of pollution concentration in rivers and water bodies at a relatively less field efforts and at a cheaper cost (Choubey, 1998; Lindell et al., 1999).

Numerous studies have accounted for the scope of optical remote sensing based analysis towards monitoring and mapping the quality of water bodies using high resolution images (Usaliand Hasmadi, 2010; Dekker et al., 2002; Schalles et al., 1998; and Ritchie and Charles, 1996). Remote sensing satellites quantifies the amount of solar radiation at several wavelength sreflected by the surface water and are capable of providing spatial and temporal data that aids the analysis of changes in the water quality necessary for adopting the strategies for better management practices (Girgin et al., 2010; Zhang et al., 2003; Jensen 2000; Ellis, 1999; Fraser, 1998; Kondratyev et al., 1998; Pattiaratchi et al., 1994).

Traditional methods for monitoring river pollutants in terms of water quality parameters (WQP's) by optical satellite data mainly rely on the spectral response of pollutants from the water body (Chavez, 1996). The spectral response towards WQP's can therefore be related by combining certain spectral bands and to generate pollution maps through regression models. The present study undertakes an attempt to monitor and map the spatial distribution of pollution concentrations corroborated by in-situ samplings at selected locations along 22 km Yamuna river stretch passing through Delhi while utilizing high resolution Geoeye2 imagery. WQP's were grouped into four classes in accordance to their chemical characteristics and their sensitivity towards four formulated spectral reflectance band combinations was explored through Multiple Linear Regression (MLR) analysis. The study demonstrates a cost-effective methodology to monitor and map pollution concentrations in urban water bodies that may find its usefulness in providing a solution for future water resources planning and management at the government level through formulating effective water quality management related policies.

STUDY AREA

The present study covers a 22 km stretch of river Yamuna passing through Delhi commencing at Wazirabad barrage and exiting at Okhla barrage (Figure 1), considered to be the worst polluted stretch amongst the entire river length. Delhi state sprawls over 1483-sq km between latitude $28^{\circ} 34' \text{ N}$ and a longitude of $77^{\circ} 07' \text{ E}$ having an elevation of 213 m above the mean sea level and experiences heavy rains particularly during the monsoon season. Water released downstream of Wazirabad barrage is the untreated or partially treated domestic and industrial wastewater along with the water transported by Haryana Irrigation Department from Western Yamuna Canal (WYC). This segment receives water from seventeen sewage drains of Delhi and also from WJC and Upper Ganga Canal via Najafgarh drain and Hindon cut canal respectively. Little contribution of water is also made in this segment by Surghat, where Ganga and Yamuna water is provided for bathing purposes. This river segment terminates into Agra Canal, which is used to augment its flow for irrigation in the states of Haryana and Uttar Pradesh.

METHODOLOGY

The methodology framed in the present study constitutes three broad workflow components i.e. 1) collection of water samples and their analysis, 2) analysis of image and its standardization, 3) MLR analysis of WQP's with spectral reflectance band combinations and WQP maps generation.

Collection of Water Samples and Their Analysis

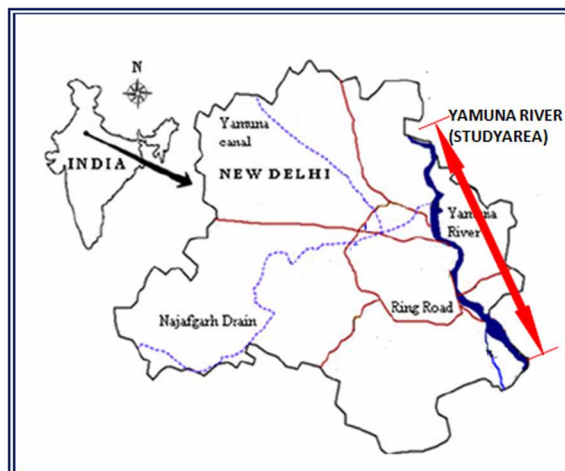
Water samples were collected from the midstream at a depth of 0.5 meters on 25th January, 2016, from the four selected sampling stations namely; Wazirabad barrage, ITO barrage, Nizamuddin bridge and Okhla barrage, where in from each sampling station 25 water samples were collected from random locations amounting to a total of 100 water samples from all four stations (Figure 2a). Grab sampling procedure was adopted for the analysis of various water quality parameters as recommended by standard methods of analysis (APHA, 1999). Samples were tested for 20 water quality parameters viz. temperature, pH, EC, turbidity, dissolved oxygen (DO), total suspended solids (TSS), total dissolved solids (TDS), turbidity, total hardness as CaCO_3 , biochemical oxygen demand (BOD), chemical oxygen demand (COD), calcium (Ca), magnesium (Mg), alkalinity, chloride (Cl), phosphate (PO_4), Sodium (Na), potassium (K), sulphate (SO_4), nitrate (NO_3) and ammonical nitrogen ($\text{NH}_3\text{-N}$). On the basis of their sensitivity towards spectral reflectance values, 15 WQP's were selected for exploring possible correlation with the reflectance values and were grouped in accordance to their chemical characteristics into four classes namely; $(\text{WQP})_{\text{organic}}$, $(\text{WQP})_{\text{inorganic}}$, $(\text{WQP})_{\text{anion}}$ and $(\text{WQP})_{\text{cation}}$. Grouping of WQP's under each class was followed as below;

- a. $(\text{WQP})_{\text{organic}} \Rightarrow \text{BOD \& TSS}$
- b. $(\text{WQP})_{\text{inorganic}} \Rightarrow \text{Cl, SO}_4, \text{NO}_3, \text{TDS}$
- c. $(\text{WQP})_{\text{anion}} \Rightarrow \text{Ca, Mg, K, Na, NH}_3 \text{ \& TDS}$
- d. $(\text{WQP})_{\text{cation}} \Rightarrow \text{non-carbonate(permanent), alkalinity and total hardness (TH)}$

Analysis of Image and its Standardization

Geoeye-2 satellite imagery of date 25th January 2016 concurrent to the water sampling date was utilized in the present study to monitor and map pollution concentrations within 22 km Yamuna river stretch. Geoeye-2 is US based satellite that collects images at a ground resolution of 34 cm (13.4

Figure 1. Location map of the study area (map not to scale)



inch) in the panchromatic or black-and-white mode and at a resolution of 1.34 m in the multispectral or color mode. Subset image of the study area was generated in Erdas Imagine software (Figure 2b).

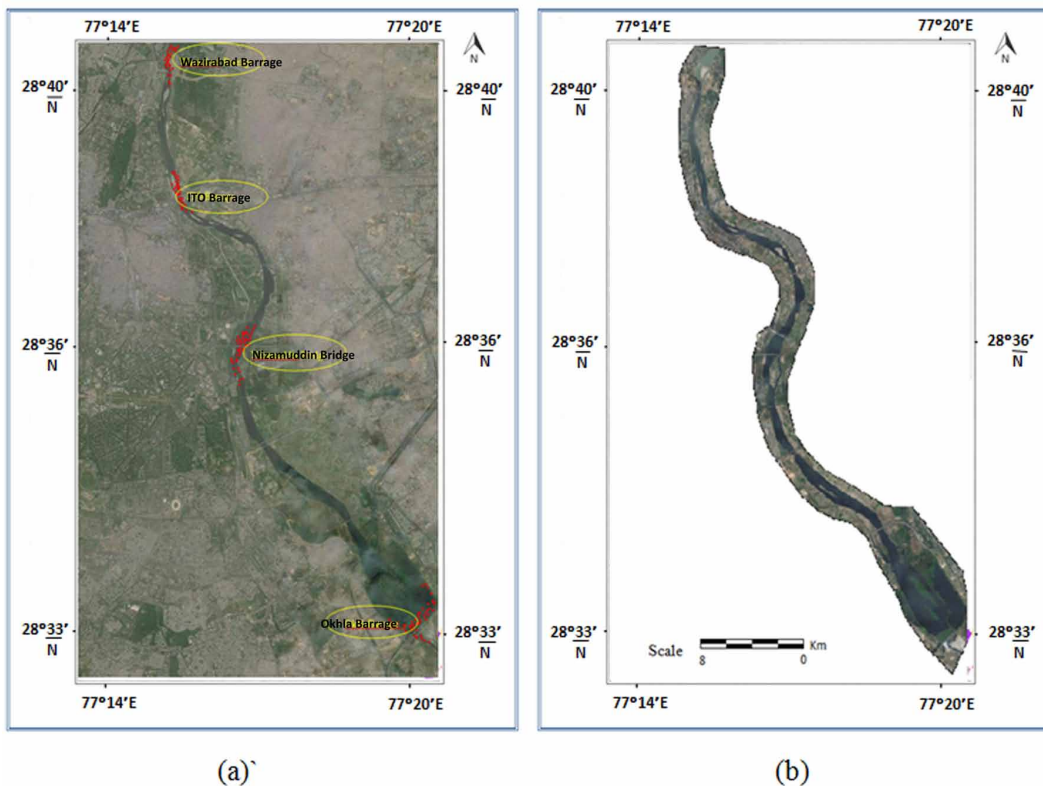
Digital numbers (DN) from the four spectral bands (i.e. RED, GREEN, BLUE and NIR bands) corresponding to the point sampling locations were extracted and transformed to radiance values and then to the reflectance values before they were subjected to the analysis and interpretation. The DN values of an image depends upon the reflectance characteristics of the specific area and also contains noise as a result of the viewing geometry of the satellite, angle of the sun's incoming radiation, atmospheric depth owing to viewing angle and design characteristics of the sensor. Conversion of DN values to radiance values removes the voltage bias and gains from the satellite sensor as described in equation (1). The radiance values are then further converted to satellite reflectance values using equation (2). This conversion takes into account the varying sun angle due to differences in latitude, season, and time of day, and the variation in the distance between the Earth and Sun (Chander and Markham, 2003; Yeom et. al, 2016).

The conversion of DN values to radiance values and finally to reflectance values corresponding to sampling locations was carried out using equations (1) and (2) respectively proposed by Chander and Markham (2003);

$$L_{\lambda} = Gain_{\lambda} \times DN_{\lambda} + offset_{\lambda} \quad (1)$$

Where; λ is the Specific spectral band of image, Near IR, Red, Green, Blue or Panchromatic, L_{λ} is the spectral radiance for band λ at the sensor's aperture ($mW/cm^2/\mu m/str$), $Gain_{\lambda}$ is the radiometric

Figure 2. (a) Geoeye-2 image of the study area depicting sampling locations marked with red dots; (b) Subset image of the study area



calibration gain ($\text{mW/cm}^2/\mu\text{m/str/DN}$) for band λ from product metadata, DN_λ is the DN value for band λ of image product and $offset_\lambda$ is the radiometric calibration offset ($\text{mW/cm}^2/\mu\text{m/str}$) for band λ obtained from product metadata.

$$\rho = \frac{\pi * L_\lambda * d^2}{E_{SUN} * \cos \theta_s} \quad (2)$$

Where; ρ_p is the unitless planetary reflectance, d is the Earth – Sun distance (astronomical units), $E_{SUN\lambda}$ is the mean solar exo-atmospheric spectral irradiances ($\text{mW/cm}^2/\mu\text{m}$), at an Earth-Sun distance of 1 astronomical unit (A.U.), θ_s is the Solar zenith angle and L_λ is the spectral radiance for band λ at the sensor's aperture ($\text{mW/cm}^2/\mu\text{m/str}$). The earth-sun distance (d) in astronomical units (A.U.) is obtained from Geoeye-2 product manual. The above process of applying corrections to DN values is called as image standardization that compensates for different sun angles at different acquisition dates as well as for the atmospheric effects. Correlations between four formulated WQP classes and reflectance values were explored for single bands as well as for several formulated band combinations. From several trials, four band combinations were selected, viz. [mean (RGB)], $\sqrt{(B/NIR)}$, [mean(RGB) $\times \sqrt{(B/R)}$] and $[(NIR-B)^2/(NIR+B)^2]$ for further comparative analysis. MLR analysis of WQP classes with single bands was discarded as low or very poor coefficient of determination was observed (i.e. $R^2 \sim 0.03$; RMSE ~ 12.46).

MLR Analysis of WQP's with Spectral Reflectance Combinations and WQP Maps Generation

Multiple Linear Regression (MLR) uses linear statistics to model the relationship between a scalar dependent variable and several independent explanatory variables. MLR analysis was performed in KY Plot statistical software between the four specified spectral band combinations and WQP classes individually for evaluating and comparing the strength of correlation. The data set utilized in the analysis was combined for all four sampling stations. The combined data set corresponds to a total of 100 samples of which 80 samples were utilized for calibration and the remaining were left over for validation purpose. Regression equations were formulated between the spectral band combination exhibiting best correlation with WQP classes for generating WQP maps in GIS framework.

RESULTS AND DISCUSSION

Water samples collected from the four sampling locations were tested for 20 water quality parameters in the laboratory and the basic statistics calculated are summarized in Table 1. Water temperatures along the river stretch showed relatively less variation with lower values concurrent with the winter foggy day ranging from 13°C to 16°C . The pH values of collected water samples ranged from 7.7 to 7.9, lying within the limit range of 6 to 9 as prescribed by Central Pollution Control Board (CPCB) India, for water quality. The abrupt increase in organic pollution content all along the river reach downstream of Wazirabad barrage resulted in biochemical oxygen demand (BOD) levels lying far beyond the stipulated standards. Also, fluctuations in dissolved oxygen (DO) level from Nil to well above saturation level reflected the presence of organic pollution load and prevalence of eutrophic conditions. TDS values showed significant variation with values ranging between 73.3 mg/l to 930 mg/l having lowest concentration level at Wazirabad ranging from 73.3 mg/l to 340 mg/l and highest at Nizamuddin bridge between 730 mg/l to 930 mg/l, exceeding far beyond the permissible limits. Altogether, the four classes of WQP, i.e. $(WQP)_{\text{organic}}$, $(WQP)_{\text{inorganic}}$, $(WQP)_{\text{anion}}$ and $(WQP)_{\text{cation}}$ were found to be at highest concentration levels exceeding the limits far beyond as prescribed by CPCB throughout the entire river section selected in the study. The abnormally high concentration levels of all WQP classes is mainly associated to domestic and industrial sewage draining into the river at

several locations commencing 250 meters downstream of Wazirabad barrage and extending up to the Okhla barrage. Further, significant water abstraction at Wazirabad barrage reduces the dilution capacity of the entire river stretch passing through Delhi.

Multiple linear regression analysis (MLR) was used to explore the relationships between the four WQP classes and spectral reflectance band combinations and the results were analyzed in terms of R^2 and RMSE values registered in Tables 2 to 5. MLR was also tested for single reflectance bands (i.e. Blue, Green, Red and Near Infrared) but exhibited poor and in some cases no correlation with WQP classes and therefore excluded in the present analysis.

Results show case positive correlation between the four opted spectral band combinations and WQP classes separately having R^2 values for calibration and validation ranging between 0.41 to 0.85 and 0.43 to 0.76 respectively within 95% confidence interval. Analysis of results testified for four spectral band combinations, $[\text{mean RGB} \times \sqrt{(B/R)}]$ showed highest correlation with all four WQP classes yielding $R^2(\text{calibration})$ values of 0.85 for inorganic class to 0.79 for anion class and $R^2(\text{validation})$ values of 0.76 for cation to 0.72 for inorganic class respectively. RMSE values were observed to be lowest for the same band combination ranging between 1.03 to 1.24 for calibration data and 1.45 to 1.69 for validation data respectively. Utilizing regression coefficients obtained from MLR analysis between $[\text{mean RGB} \times \sqrt{(B/R)}]$ and four WQP classes, numeric values of individual WQP class (i.e. $(WQP)_{\text{organic}}$, $(WQP)_{\text{inorganic}}$, $(WQP)_{\text{anion}}$ and $(WQP)_{\text{cation}}$) were computed and a simple regression analysis was performed to obtain linear equations used for generating WQP maps in GIS framework. The regression equations obtained are provided below (i.e., Equations 3 to 6) and scatter plots were drawn as shown in Figure 3a to 3d;

1. $(WQP)_{\text{organic}} = 1.163 \times [\text{mean RGB} \times \sqrt{B/R}] - 0.03711$ (3)
2. $(WQP)_{\text{inorganic}} = 1.113 \times [\text{mean RGB} \times \sqrt{B/R}] - 0.02411$ (4)
3. $(WQP)_{\text{anion}} = 1.089 \times [\text{mean RGB} \times \sqrt{B/R}] - 0.0219$ (5)

Table 1. Range, mean and standard deviation of WQP's at sampling locations

S. No.	Parameters (mg/l)	Range Min - Max	Mean	SD
1	BOD	4.55 - 45	29.32446	14.41
2	COD	14.17 - 150	100.662	48.93
3	TSS	160 - 250	209.77	23.51
4	Cl	81.45 - 320	198.34	88.09
5	SO4	47 - 119	79.27	21.31
6	NO3	0.82 - 1.32	1.074	0.146
7	TDS	73.3 - 930	384.69	278.99
8	Ca	34.7 - 65	49.03	9.07
9	Mg	58.37 - 121	86.35	23.82
10	Potassium	8.5 - 29.6	17.02	5.75
11	Sodium	124 - 266	197.67	49.72
12	NH3-N	8.3 - 41.4	21.93	9.97
13	Non Carbonate (P)	240 - 430	314.97	67.23
14	Alkalinity	90 - 240	165.55	51.03
15	Total Hardness	331.5 - 661.7	481.97	119.75

Table 2. Multiple linear regression coefficients and R^2 values for [mean RGB $\times \sqrt{(B/R)}$] image with combined data of all sampling locations

Multiple Regression Equation $Y = b_0 + b_1 \cdot X_1 + b_2 \cdot X_2 + b_3 \cdot X_3 + \dots$ Where $Y = [\text{mean RGB} \times \sqrt{(B/R)}]$ Data distribution: 80% Correlation and 20% Validation					
WQP class	Reg. Coeff	R^2 Calibration	RMSE Calibration	R^2 Validation	RMSE Validation
(WQP) _{organic} = BOD, TSS	$b_0 = 0.4852$ $b_1 = -0.0023$ $b_2 = -0.00094$	0.82	1.12	0.74	1.64
(WQP) _{Inorganic} = Cl, SO ₄ , NO ₃ , TDS	$b_0 = 0.5481$ $b_1 = -0.00012$ $b_2 = -0.00213$ $b_3 = -0.07229$ $b_4 = -0.00012$	0.85	1.03	0.72	1.45
(WQP) _{anion} = Ca, Mg, K, Na, NH ₃ , TDS	$b_0 = 0.38335$ $b_1 = -0.00024$ $b_2 = 0.003243$ $b_3 = -0.00017$ $b_4 = -0.00199$ $b_5 = -0.00151$ $b_6 = -1.1 \times 10^{-6}$	0.79	1.24	0.74	1.58
(WQP) _{cation} = Carbonate(p), Alkalinity, TH	$b_0 = 0.34215$ $b_1 = -0.00052$ $b_2 = -0.00208$ $b_3 = 0.00082$	0.83	1.11	0.76	1.69

Table 3. Multiple linear regression coefficients and R^2 values for [mean RGB] image with combined data of all sampling locations

Multiple Regression Equation $Y = b_0 + b_1 \cdot X_1 + b_2 \cdot X_2 + b_3 \cdot X_3 + \dots$ Where $Y = [\text{mean RGB}]$ Data distribution: 80% Correlation and 20% Validation					
WQP class	Reg. Coeff	R^2 Calibration	RMSE Calibration	R^2 Validation	RMSE Validation
(WQP) _{organic} = BOD, TSS	$b_0 = 0.585234$ $b_1 = 0.585234$ $b_2 = -0.002961$	0.58	2.58	0.54	2.81
(WQP) _{Inorganic} = Cl, SO ₄ , NO ₃ , TDS	$b_0 = 0.868205$ $b_1 = -0.043352$ $b_2 = 0.022571$ $b_3 = -0.022144$ $b_4 = -0.002845$	0.56	2.74	0.51	3.06
(WQP) _{anion} = Ca, Mg, K, Na, NH ₃ , TDS	$b_0 = 0.625562$ $b_1 = -0.001536$ $b_2 = 0.0044139$ $b_3 = -0.002198$ $b_4 = -0.042991$ $b_5 = -0.042341$ $b_6 = -2.1 \times 10^{-6}$	0.48	3.26	0.44	2.73
(WQP) _{cation} = Carbonate(p), Alkalinity, TH	$b_0 = 0.346374$ $b_1 = -0.002254$ $b_2 = -0.004388$ $b_3 = 0.001261$	0.63	1.94	0.59	2.42

Table 4. Multiple linear regression coefficients and R² values for $\sqrt{(B/NIR)}$ image with combined data of all sampling locations

Multiple Regression Equation $Y = b_0 + b_1 \cdot X_1 + b_2 \cdot X_2 + b_3 \cdot X_3 + \dots$ Where $Y = \sqrt{(B/NIR)}$ Data distribution: 80% Correlation and 20% Validation					
WQP class	Reg. Coeff	R ² Calibration	RMSE Calibration	R ² Validation	RMSE Validation
(WQP) _{organic} = BOD, TSS	$b_0 = 0.492534$ $b_1 = 0.076341$ $b_2 = -0.005811$	0.57	2.52	0.54	3.16
(WQP) _{Inorganic} = Cl, SO ₄ , NO ₃ , TDS	$b_0 = 0.632415$ $b_1 = -0.033521$ $b_2 = 0.053461$ $b_3 = -0.046124$ $b_4 = -0.004254$	0.49	2.76	0.47	2.84
(WQP) _{anion} = Ca, Mg, K, Na, NH ₃ , TDS	$b_0 = 0.834522$ $b_1 = -0.004258$ $b_2 = 0.0024659$ $b_3 = -0.007546$ $b_4 = -0.065241$ $b_5 = -0.056426$ $b_6 = -0.0024745$	0.45	3.58	0.43	3.91
(WQP) _{cation} = Carbonate(p), Alkalinity, TH	$b_0 = 0.565214$ $b_1 = -0.005412$ $b_2 = -0.012358$ $b_3 = 0.007852$	0.61	2.14	0.59	2.24

Table 5. Multiple linear regression coefficients and R² values for $[(B-NIR)^2/(B+NIR)^2]$ image with combined data of all sampling locations

Multiple Regression Equation $Y = b_0 + b_1 \cdot X_1 + b_2 \cdot X_2 + b_3 \cdot X_3 + \dots$ Where $Y = [(B-NIR)^2/(B+NIR)^2]$ Data distribution: 80% Correlation and 20% Validation					
WQP class	Reg. Coeff	R ² Calibration	RMSE Calibration	R ² Validation	RMSE Validation
(WQP) _{organic} = BOD, TSS	$b_0 = 0.125834$ $b_1 = 0.093341$ $b_2 = -0.012851$	0.59	3.14	0.61	3.58
(WQP) _{Inorganic} = Cl, SO ₄ , NO ₃ , TDS	$b_0 = 0.562415$ $b_1 = -0.035273$ $b_2 = 0.068241$ $b_3 = -0.025241$ $b_4 = -0.01286$	0.54	2.68	0.51	3.24
(WQP) _{anion} = Ca, Mg, K, Na, NH ₃ , TDS	$b_0 = 0.562544$ $b_1 = 0.087258$ $b_2 = 0.011589$ $b_3 = 0.046851$ $b_4 = -0.046852$ $b_5 = -0.055616$ $b_6 = -0.0142515$	0.41	4.67	0.48	4.95
(WQP) _{cation} = Carbonate(p), Alkalinity, TH	$b_0 = 0.821144$ $b_1 = -0.02682$ $b_2 = 0.025678$ $b_3 = 0.045772$	0.52	2.84	0.45	3.57

$$4. (WQP)_{cation} = 1.489 \times [\text{mean RGB} \times \sqrt{B/R}] - 0.03389 \quad (6)$$

WQP maps (Figure 4) clearly demonstrates the concentration levels of WQP's in terms of organic, inorganic, anion and cation to be far above the permissible limits as prescribed by CPCB throughout the entire length of the river reach. At the upstream and some distance downstream of Wazirabad barrage, water quality reflected in the maps reveals low or minimum pollution concentrations due to the fact that there are no point or non-point sources of waste water discharge into the river and is testified with the light color tones in WQP maps. Barely 250 m downstream of Wazirabad barrage, Najafgarh drain joins the river stream thereby causing the abrupt rise of pollution concentrations, since drain carries industrial effluents as well as domestic sewage and is considered as the starting point of river pollution. The pollution level increases far beyond the permissible limits as more and more major drains (approx. 25 drains) carrying industrial effluent as well as domestic sewage discharging into the river as evidenced by darker colour tones along the entire river reach in the WQP maps. The above observation can also be attributed to the fact that the point sampling and analysis was carried out during the winter season (i.e. 25th January 2016) wherein the pollution concentrations are at the highest levels due to negligibly low river flows.

CONCLUSION

The present study utilizes high resolution Geosye-2 imagery for monitoring and mapping the pollution levels of 22 km Yamuna river stretch passing through Delhi. A total of 100 water samples collected were tested for 20 water quality parameters and based on the chemical characteristics were grouped into four classes namely; (WQP)_{organic}, (WQP)_{inorganic}, (WQP)_{anion} and (WQP)_{cation} respectively. On the basis of several trials conducted with single bands as well as band combinations, four spectral reflectance band combinations formulated were finalized that exhibited positive correlation, namely; [mean (RGB)], $\sqrt{(B/NIR)}$, $[\text{mean}(\text{RGB}) \times \sqrt{(B/R)}]$ and $[(NIR-B)^2/(NIR+B)^2]$ for performing MLR analysis individually with the four WQP classes. It was observed that [mean(RGB)] band combination depicted lowest correlation with all four WQP classes yielding R² values for calibration and validation data sets in the range of 0.48 to 0.63 (RMSE: 1.94 to 3.26) and 0.44 to 0.59 (RMSE: 2.42 to 3.06)

Figure 3. Scatter plots of $[\text{mean RGB} \times \sqrt{(B/R)}]$ values with four WQP's classes

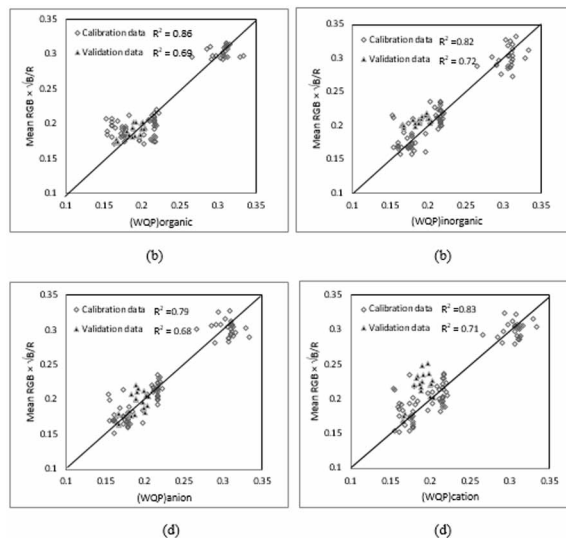
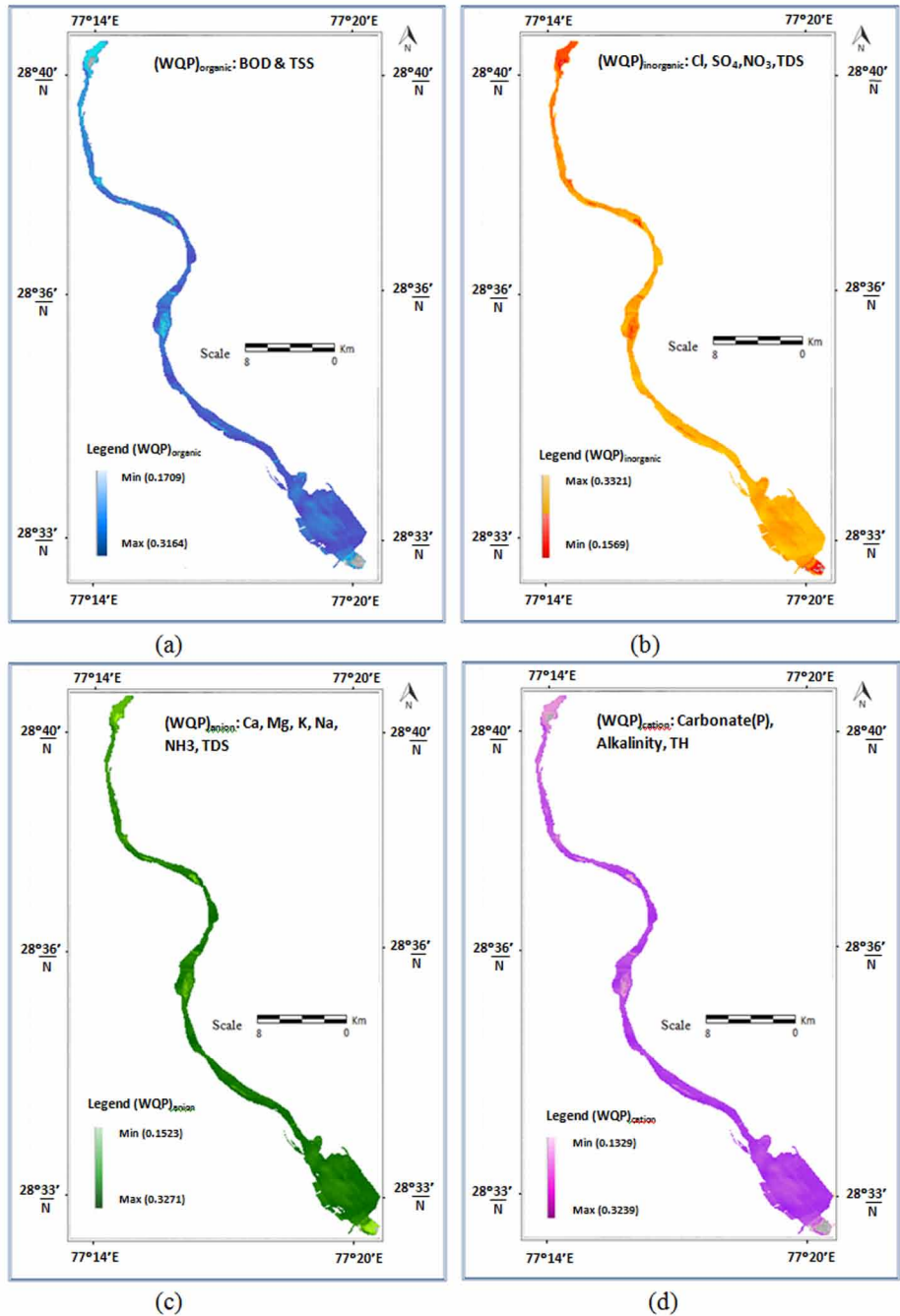


Figure 4. WQP maps of the study area



respectively. Whereas, $[\text{mean}(\text{RGB}) \times \sqrt{(\text{B/R})}]$ indicated relatively stronger correlation with all four WQP classes with R^2 for calibration and validation data sets ranging from 0.79 to 0.85 (RMSE: 1.03 to 1.24) and 0.72 to 0.76 (RMSE: 1.45 to 1.69) respectively. WQP maps generated corresponding to $[\text{mean}(\text{RGB}) \times \sqrt{(\text{B/R})}]$ and WQP classes apparently testified lowest pollution concentration

levels in terms of all four WQP classes at Wazirabad sampling station as depicted by the light colour tones and abnormally high concentration levels of all four WQP classes all along the river reach as exhibited by the darker tones. The study, however demonstrates the technical potential of Geoeye-2 imagery for monitoring and mapping pollution levels in the water bodies that may find its usefulness for better planning and management of surface water resources.

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Athar Hussain is working as an Associate Professor in Department of Civil Engineering and Head, Environmental Engineering Department at Ch. B. P. Govt. College of Engineering, New Delhi. Dr. Hussain completed his PhD in Environmental Engineering from IIT Roorkee in year 2007. During the research period Dr. Hussain worked as SRF in CSIR and MoEF sponsored projects till Dec 2006. Thereafter Dr. Hussain started his career as a lecturer in civil engineering department at Dr. B.R. Ambedkar National Institute of Technology, Jalandhar till May 2007. Thereafter he joined as an Assistant Director with Environment Management Group at National Productivity Council, New Delhi an autonomous body under Ministry of Commerce & Industry and worked there till Oct. 2010. During his tenure he was involved in various national level and policy level projects sponsored from reputed government agencies and organizations such as MoEF, CPCB, SPCBs etc. in the field of environmental engineering. Thereafter he worked as an Assistant Professor, civil engineering department in School of Engineering at Gautam Buddha University till December 2015. Thereafter he joined as Associate Professor in Department of Civil Engineering at Ch. B. P. Govt. College of Engineering, New Delhi. His research area includes industrial wastewater treatment and solid waste management with special focus on anaerobic treatment of wastewater and solid waste. He is supervising many PhD Research Scholars in diversified areas of environmental engineering. Dr. Hussain has more than 15 years of experience in the area of teaching, research and consultancy. He has also guided more than 60 MTech dissertations. He has published more than 60 papers in international referred journals and international conferences. Dr. Hussain is also serving as a reviewer for many International journals like Taylor and Francis, Inderscience Publishers, Elsevier Science. etc. He is also an editorial board member of many journals, some of which include Elsevier science and Inderscience publications. He is also an Associate Member of The Institution of Engineers (IE) India, Member of Institution of valuers (IV) N. Delhi, India and Member of Indian Society of Wind Engineering, India. He has also handled various national and policy level projects with government and private organizations.

Saif Said obtained a PhD from Indian Institute of Technology (IIT) Roorkee and has worked for nine years (09) as assistant professor in the Department of Civil Engineering, Jamia Millia Islamia University, New Delhi, and served at Thapar Institute of Engineering and Technology presently known as Thapar University, Patiala in Punjab as assistant professor for brief period. At present, he is an associate professor in Civil Engineering Department, Aligarh Muslim University (AMU), Aligarh. He has more than nine years of teaching and research experience. He is Fellow of the Indian Water Resource Society (IWRS) and life member of The Institution of Engineers (IE), India. Dr. Said is actively associated with teaching with Hydraulic engineering courses and research as well as consultancy activities in various aspects of Water Resources Engineering. He has published more than 30 research papers in international and national journals of repute as well as conference proceedings.